1: Prepare the state $|0\rangle$ across the other registers.
2: **for** $k = 1 \dots q$ **do**
3: Apply V_i on registers $B_1 C_1 \dots B_{k-1} C_{k-1} B_k$.
4: Apply U_X to registers $B_k C_k$.
5: **end for**
6: Perform POVM on registers $B_1 C_1 \dots B_q C_q$ and output result.